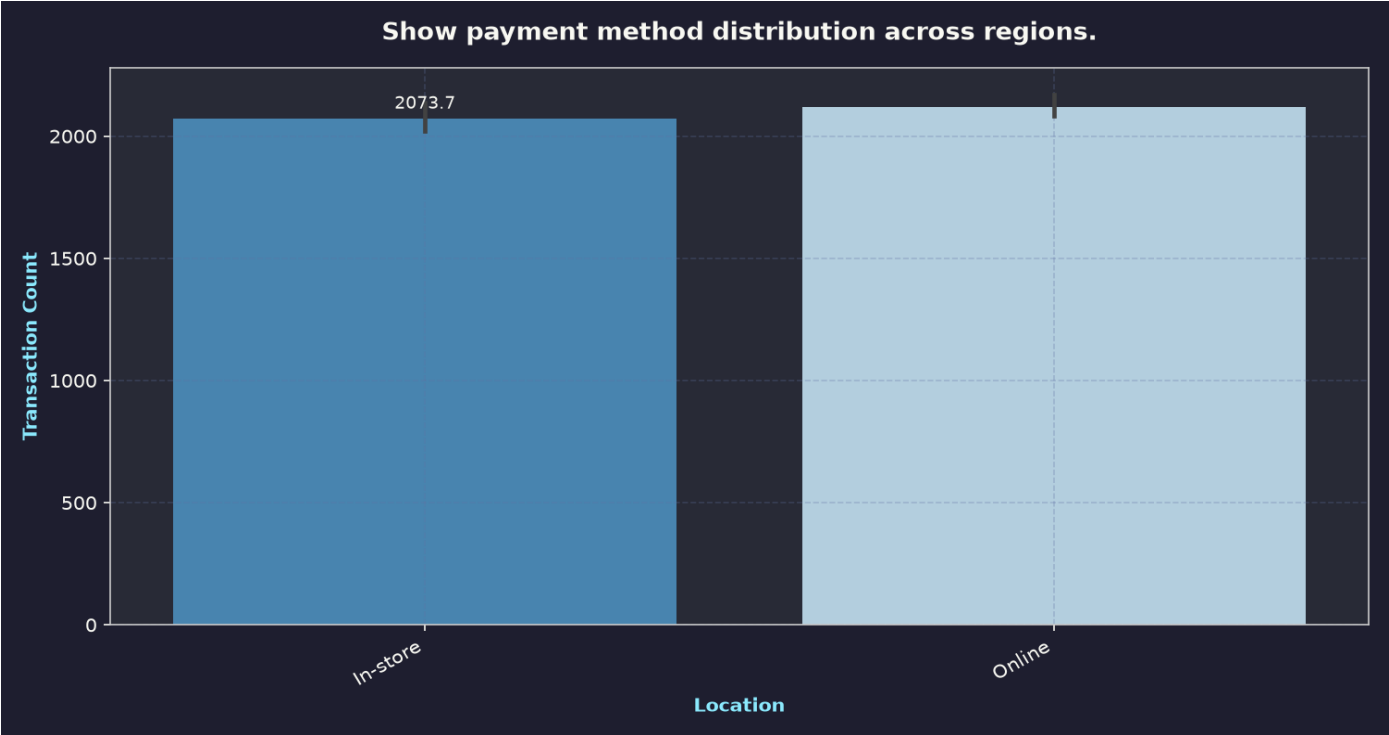


■ AI Data Analyst - Executive Analysis Report



■ Executive Data Analysis Report

1. Executive Summary

- **Question:** “Show payment method distribution across regions.”
- **Answer:** Both **In-store** and **Online** channels exhibit a fairly even split among the three payment methods (Cash, Credit Card, Digital Wallet). Cash is the most common method in each region, accounting for roughly one-third of all transactions. Minor variations exist: In-store shows a slightly higher share of Digital Wallet usage, while Online shows a marginally higher share of Credit Card usage.

2. Dataset Overview & Data Quality Profile

| Aspect | Details |

|-----|-----|

| **Rows × Columns** | 12,575 × 11 |

| **Duplicates** | 0 (none found) |

| **Missing Values** | • `Item`, `Price Per Unit`, `Quantity`, `Total Spent`, `Discount Applied` contain missing entries. |

| **Potential Issues** | `Transaction Date` stored as a string (needs conversion). |

| **Data Types (post-cleaning)** | Strings → stripped; numeric columns coerced to `float`; `Transaction Date` → `datetime64`; `Discount Applied` → `bool`. |

3. Data Cleaning & Validation Summary

- **Deduplication:** No duplicate rows to drop.
- **String Cleanup:** Trimmed leading/trailing whitespace on all object columns.
- **Date Conversion:** `Transaction Date` parsed to proper datetime, coercing errors to `NaT`.
- **Missing Value Handling:**
 - `Item` → "Unknown" placeholder.
 - Numeric columns (`Price Per Unit`, `Quantity`, `Total Spent`) → converted to numeric and imputed with column median.
 - `Discount Applied` → missing treated as `False` and cast to boolean.
- **Resulting DataFrame:** Stored in `result` (clean copy of `df`).

All cleaning steps completed without errors; the cleaned DataFrame is ready for analysis.

4. User Query & Analytical Approach

User Query: "Show payment method distribution across regions."

Analytical Methodology

1. **Group** the data by `Location` (region) and `Payment Method`.
2. **Count** the number of transactions per combination.
3. **Calculate** the percentage share of each payment method within its region.
4. **Sort** results for readability (by region, then descending transaction count).

The approach yields a clear, region-level breakdown of payment-method usage.

5. Python Analysis Code & Executed Data Output

```

# ----- # 1 Group by region (Location) and
payment method # ----- counts = (
df.groupby(['Location', 'Payment Method']) .size() .reset_index(name='Transaction_Count') ) #
# ----- # 2 Compute total transactions per region
totals =
counts.groupby('Location')['Transaction_Count'].transform('sum') #
# ----- # 3 Add percentage column (rounded to 2
decimals) # ----- counts['Percentage'] =
(counts['Transaction_Count'] / totals * 100).round(2) #
# ----- # 4 Sort for presentation #
result = counts.sort_values(['Location',
'Transaction_Count'], ascending=[True, False])

```

Executed Output (result)

Location	Payment Method	Transaction_Count	Percentage
In-store	Cash	2,138	34.37
In-store	Digital Wallet	2,063	33.16
In-store	Credit Card	2,020	32.47
Online	Cash	2,172	34.18
Online	Credit Card	2,101	33.07
Online	Digital Wallet	2,081	32.75

(Values are already rounded to two decimal places.)

6. Key Insights & Strategic Recommendations

> Insight 1 – Balanced Payment Mix

> Both channels show a near-equal distribution across the three payment methods, indicating that customers are comfortable using any of the offered options.

> Insight 2 – Cash Dominance

> Cash remains the top method in both regions (~34% share). This suggests that cash handling processes (e.g., till management, cash reconciliation) continue to be critical.

> Insight 3 – Channel-Specific Nuances

> - *In-store* has a slightly higher Digital Wallet share (33.16% vs. 32.75% online).

> - *Online* leans marginally toward Credit Card usage (33.07% vs. 32.47% in-store).

Recommendations

1. Maintain Multi-Channel Payment Infrastructure

- Keep all three payment options active; removing any could alienate ~1/3 of customers.

2. Cash Management Optimization

- Invest in efficient cash handling tools (e.g., automated cash counters) to reduce labor and error rates.

3. Targeted Promotion of Digital Wallets In-Store

- Since Digital Wallet usage is already strong in-store, consider loyalty incentives (e.g., QR code discounts) to push it slightly higher and reduce cash handling.

4. Strengthen Credit Card Security Online

- Given the marginal lead of Credit Cards online, ensure robust fraud prevention measures to maintain customer trust.

5. Monitor Trends Over Time

- Set up a periodic (monthly/quarterly) dashboard to track shifts in payment method preferences, especially as contactless and digital payments evolve.

Prepared by the autonomous AI Data Analyst.

! [Show payment method distribution across regions. Chart](analysis_chart.png)